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Qv Help Search

```
> print(vehicle_data)
Vehicle_ID Engine_Size Horsepower Fuel_Efficiency Top_Speed Safety_Rating
1          V1          1.5          110             18       180          4
2          V2          2.0          150             15       200          5
3          V3          3.0          250             12       250          5
4          V4          2.5          200             14       220          4
5          V5          1.8          130             17       190          3
>
> library(ggplot2)
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # -----
> # Q1: Violin Plot - Fuel Efficiency by Safety Rating
> # -----
> vehicle_data$Safety_Rating <- as.factor(vehicle_data$Safety_Rating)
>
> new_window()
> print(
+   ggplot(vehicle_data, aes(x = Safety_Rating, y = Fuel_Efficiency, fill = Safety_Rating)) +
+     geom_violin(trim = FALSE, alpha = 0.7) +
+     geom_point(size = 3, color = "black") +
+     scale_fill_manual(values = c("3" = "salmon", "4" = "skyblue", "5" = "lightgreen")) +
+     labs(title = "Violin Plot: Fuel Efficiency by Safety Rating",
+          x = "Safety Rating",
+          y = "Fuel Efficiency (km/l)",
+          fill = "Safety Rating") +
+     theme_minimal()
+ )
Warning message:
Groups with fewer than two datapoints have been dropped.
i Set `drop = FALSE` to consider such groups for position adjustment purposes.
>
> # -----
> # Q2: Scatter Plot - Horsepower vs Top Speed (color by Engine Size)
> # -----
> new_window()
> print(
+   ggplot(vehicle_data, aes(x = Horsepower, y = Top_Speed, color = Engine_Size)) +
+     geom_point(size = 5) +
+     geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
+     scale_color_gradient(low = "blue", high = "red") +
+     geom_text(aes(label = Vehicle_ID), vjust = -1, size = 3.5) +
+     labs(title = "Horsepower vs Top Speed (Color = Engine Size)",
+          x = "Horsepower",
+          y = "Top Speed (km/h)",
+          color = "Engine Size (L)") +
+     theme_minimal()
```

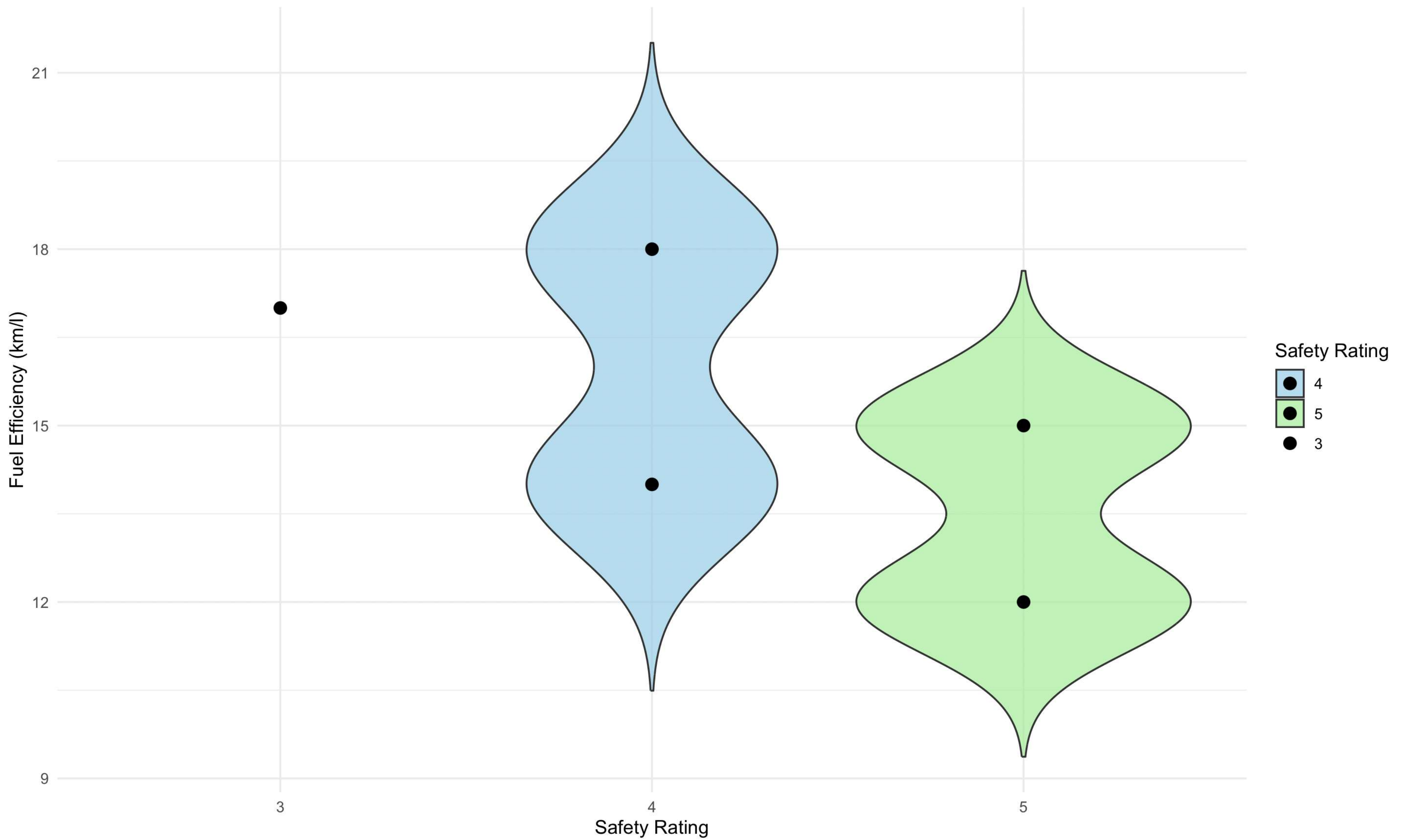


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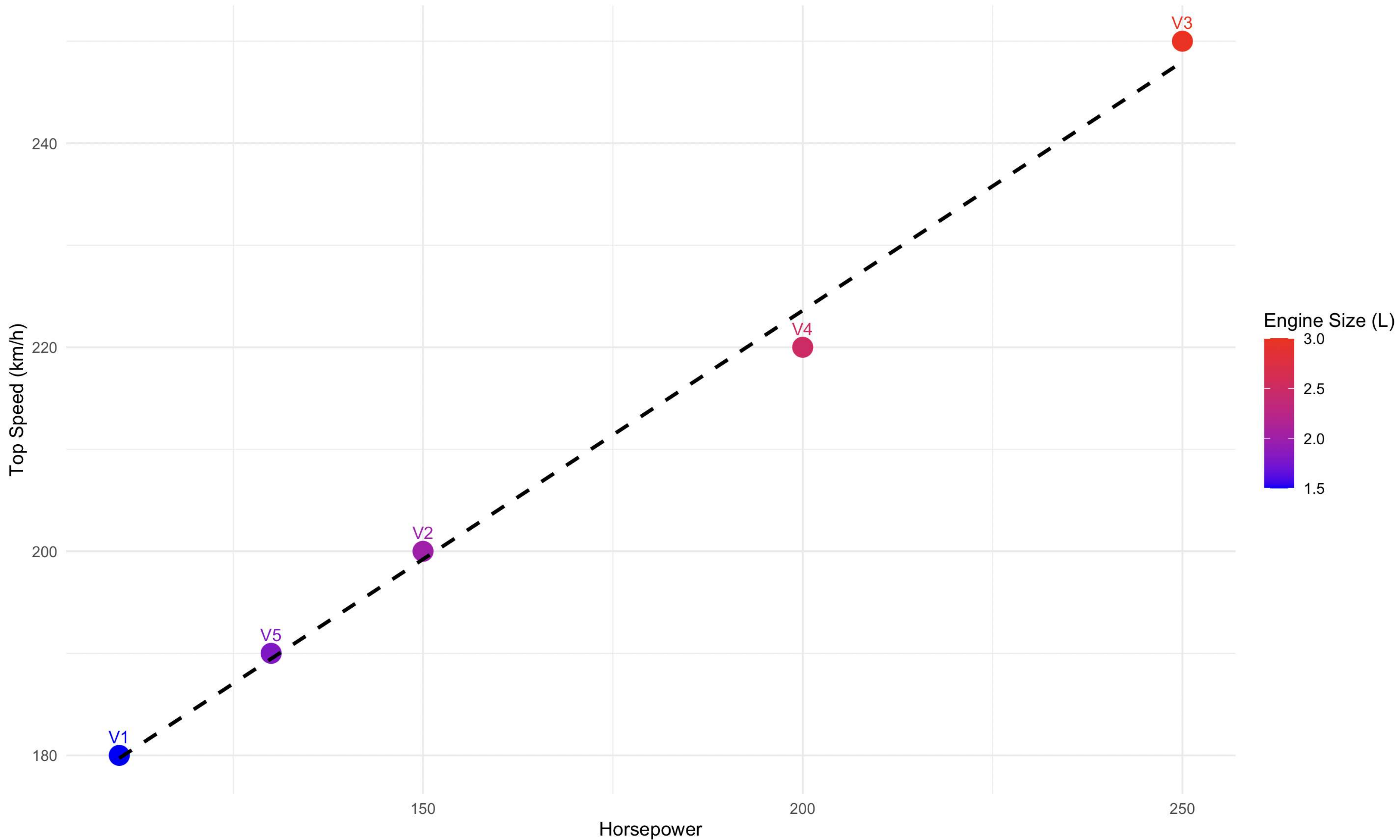
Qv Help Search

```
> # Q2: Scatter Plot - Horsepower vs Top Speed (color by Engine Size)
> # -----
> new_window()
> print(
+   ggplot(vehicle_data, aes(x = Horsepower, y = Top_Speed, color = Engine_Size)) +
+     geom_point(size = 5) +
+     geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
+     scale_color_gradient(low = "blue", high = "red") +
+     geom_text(aes(label = Vehicle_ID), vjust = -1, size = 3.5) +
+     labs(title = "Horsepower vs Top Speed (Color = Engine Size)",
+           x = "Horsepower",
+           y = "Top Speed (km/h)",
+           color = "Engine Size (L)") +
+     theme_minimal()
+ )
`geom_smooth()` using formula = 'y ~ x'
>
> # -----
> # Q3: Correlation Heatmap (No extra packages needed)
> # -----
> # Convert Safety_Rating back to numeric for correlation
> vehicle_data$Safety_Rating <- as.numeric(as.character(vehicle_data$Safety_Rating))
>
> num_data <- vehicle_data[, c("Engine_Size", "Horsepower", "Fuel_Efficiency",
+                               "Top_Speed", "Safety_Rating")]
> corr_matrix <- round(cor(num_data), 2)
> print(corr_matrix)
      Engine_Size Horsepower Fuel_Efficiency Top_Speed Safety_Rating
Engine_Size      1.00      1.00         -0.98      0.99          0.52
Horsepower       1.00      1.00         -0.97      1.00          0.54
Fuel_Efficiency  -0.98     -0.97          1.00     -0.97         -0.65
Top_Speed        0.99      1.00         -0.97      1.00          0.56
Safety_Rating    0.52      0.54         -0.65      0.56          1.00
>
> # Melt using base R (no reshape2 needed)
> corr_melted <- data.frame(
+   Var1 = rep(rownames(corr_matrix), times = ncol(corr_matrix)),
+   Var2 = rep(colnames(corr_matrix), each = nrow(corr_matrix)),
+   value = as.vector(corr_matrix)
+ )
>
> new_window()
> print(
+   ggplot(corr_melted, aes(x = Var1, y = Var2, fill = value)) +
+     geom_tile(color = "white") +
+     geom_text(aes(label = value), size = 4, fontface = "bold") +
+     scale_fill_gradient2(low = "blue", mid = "white", high = "red",
+                           midpoint = 0, limit = c(-1, 1)) +
+     labs(title = "Correlation Heatmap - Vehicle Performance Variables",
+           x = "", y = "", fill = "Correlation") +
+     theme_minimal() +
+     theme(axis.text.x = element_text(angle = 45, hjust = 1))
+ )
> |
```

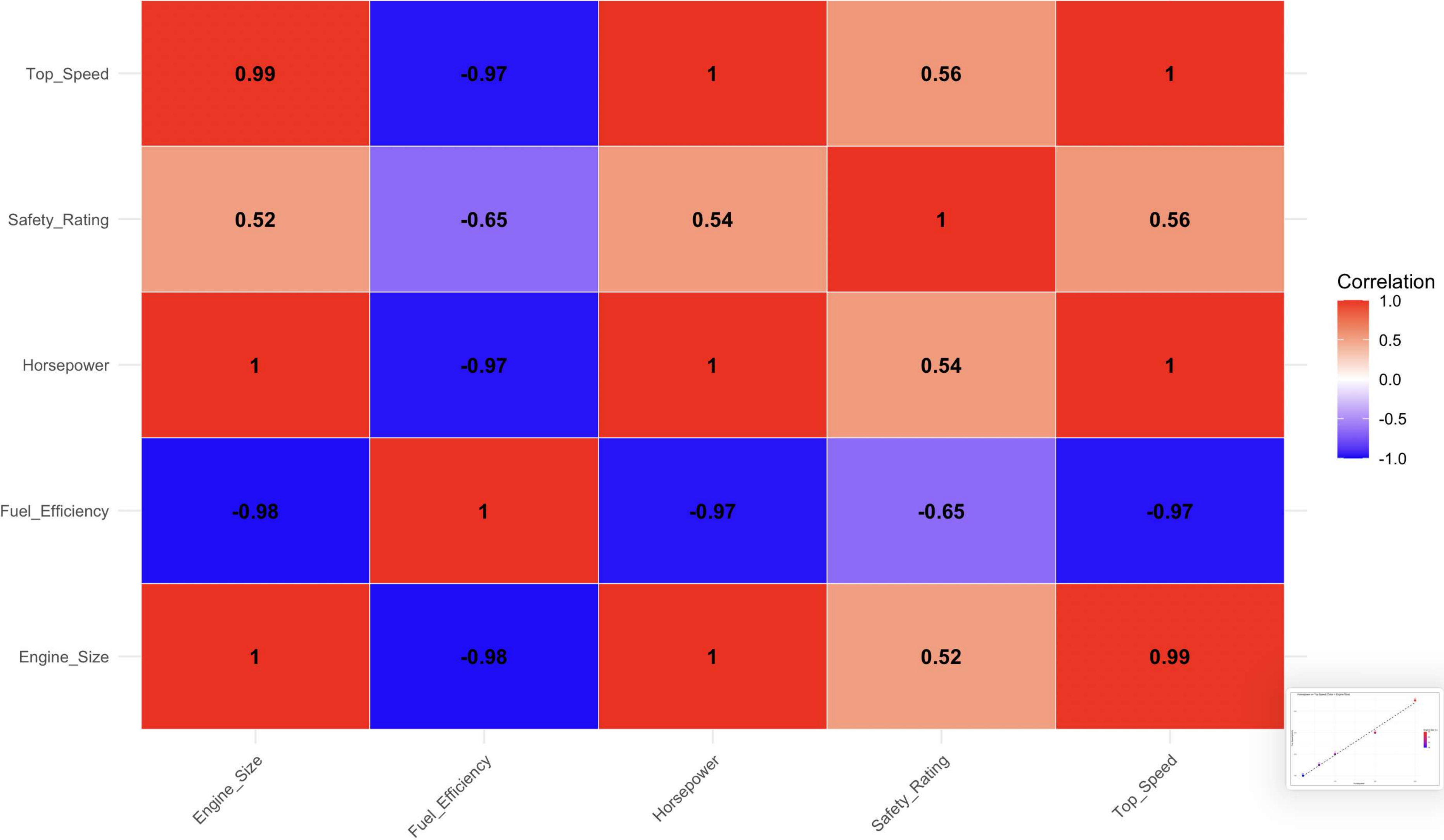
Violin Plot: Fuel Efficiency by Safety Rating



Horsepower vs Top Speed (Color = Engine Size)



Correlation Heatmap - Vehicle Performance Variables



Dashboard

Layout

Default

Phone

Device Preview

Size

Automatic

Sheets

Sheet 1

Sheet 2

Objects

Horizontal Container

Vertical Container

Text

Extension

Pulse Metric

Image

Blank

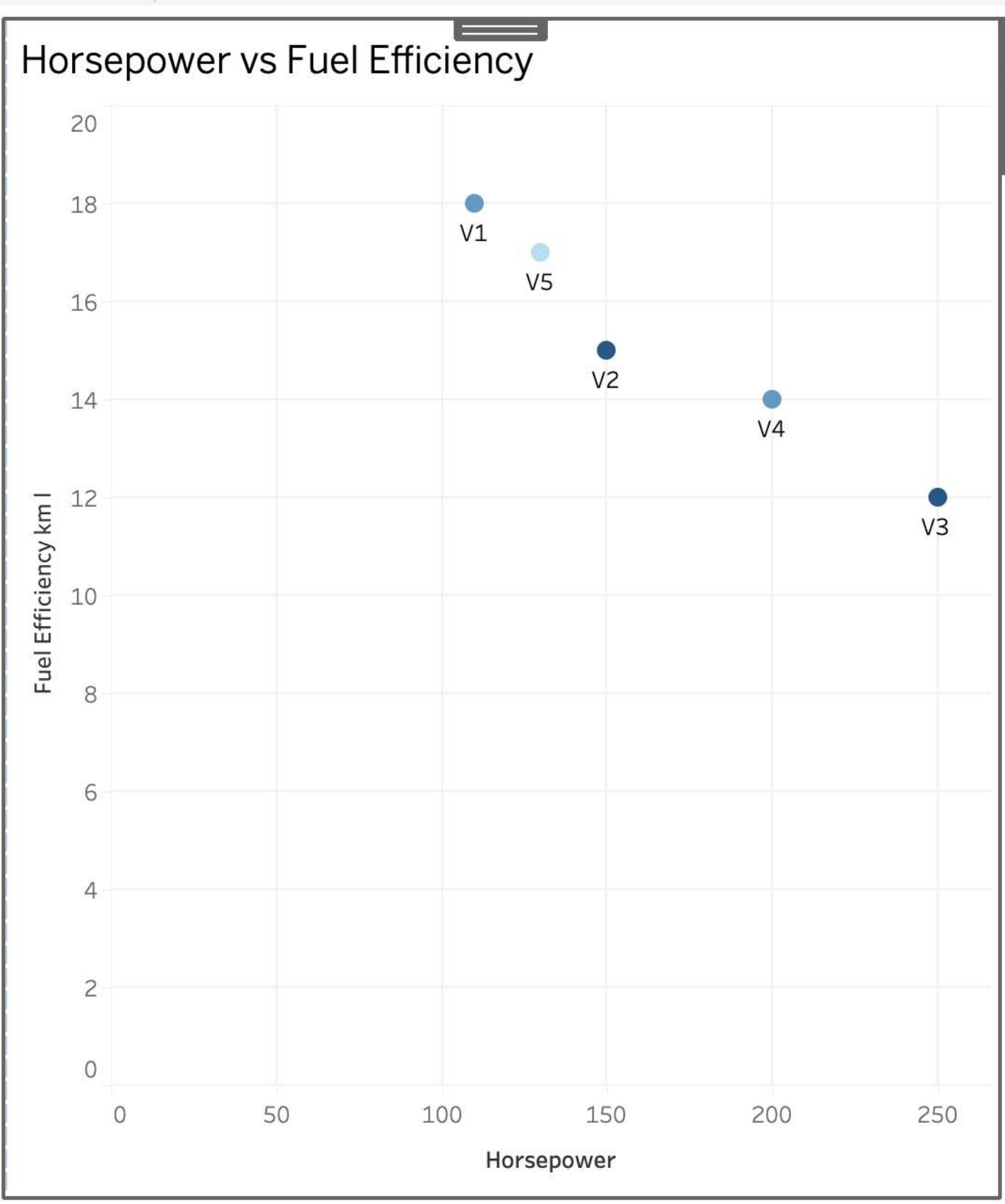
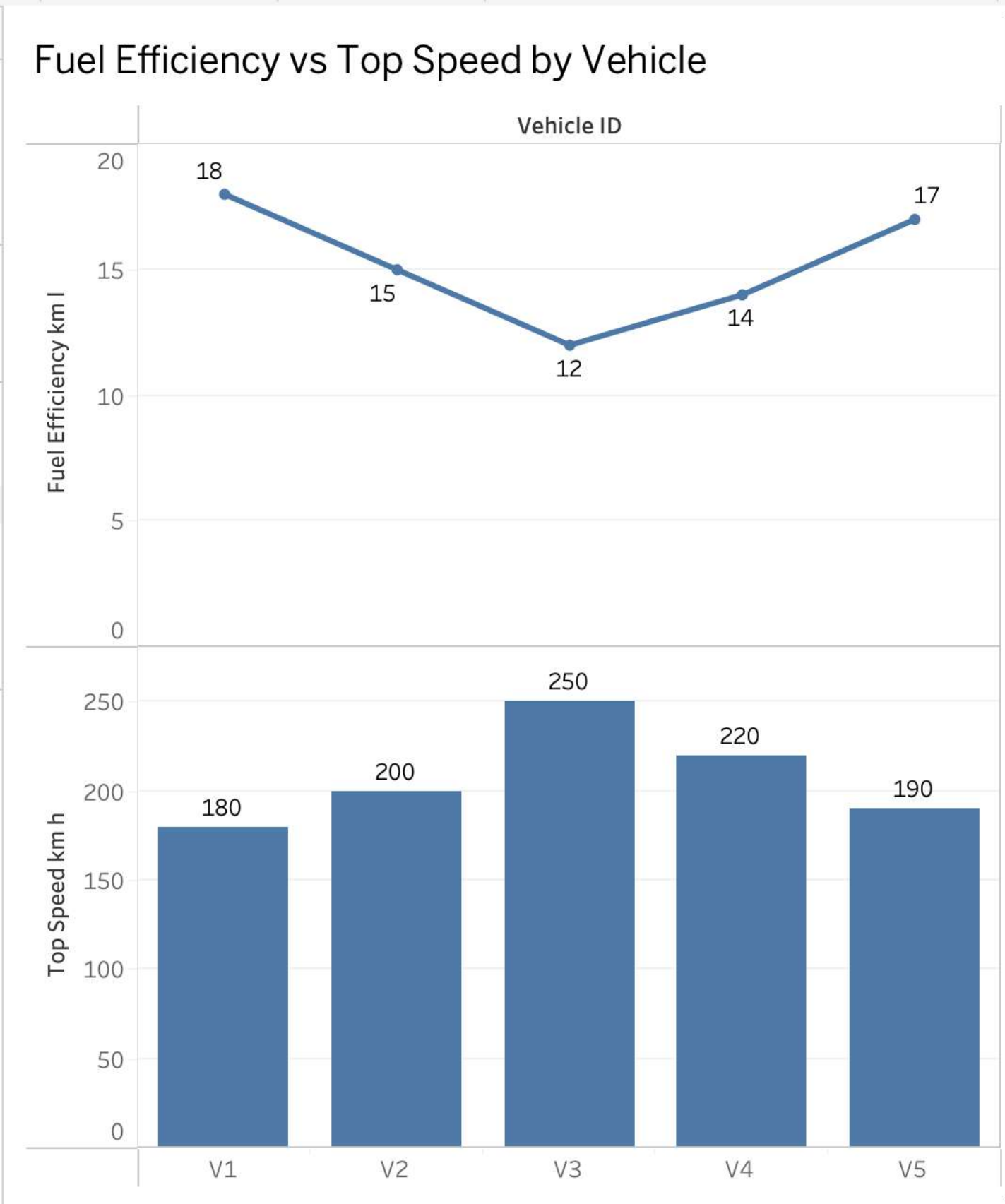
Workflow

Web Page

Tiled

Floating

Show dashboard title



Safety Rating

5